

E11 Ignition-Extinction curve

Sunday, July 26, 2026

1:26 PM

Why Are We Learning This?

An ignition-extinction curve tells us how a non-isothermal CSTR responds as the feed temperature changes. Although a reactor may have multiple possible steady states, it will not randomly jump between them. Instead, the reactor's operating temperature depends on where it was operating previously and how the operating conditions change.

Understanding ignition-extinction behavior helps engineers answer questions such as: Will the Reactor Heat Up? Will the Reactor Cool Down? Will the Reactor Jump to a New Steady State? Is Runaway Possible?

These questions are critical during reactor startup, shutdown, and normal operation.

Course Roadmap

Introduction to CRE



General Balance Equation



Common Reactor Models
(CSTR, Batch, PFR)



Reaction Progress
(Conversion)



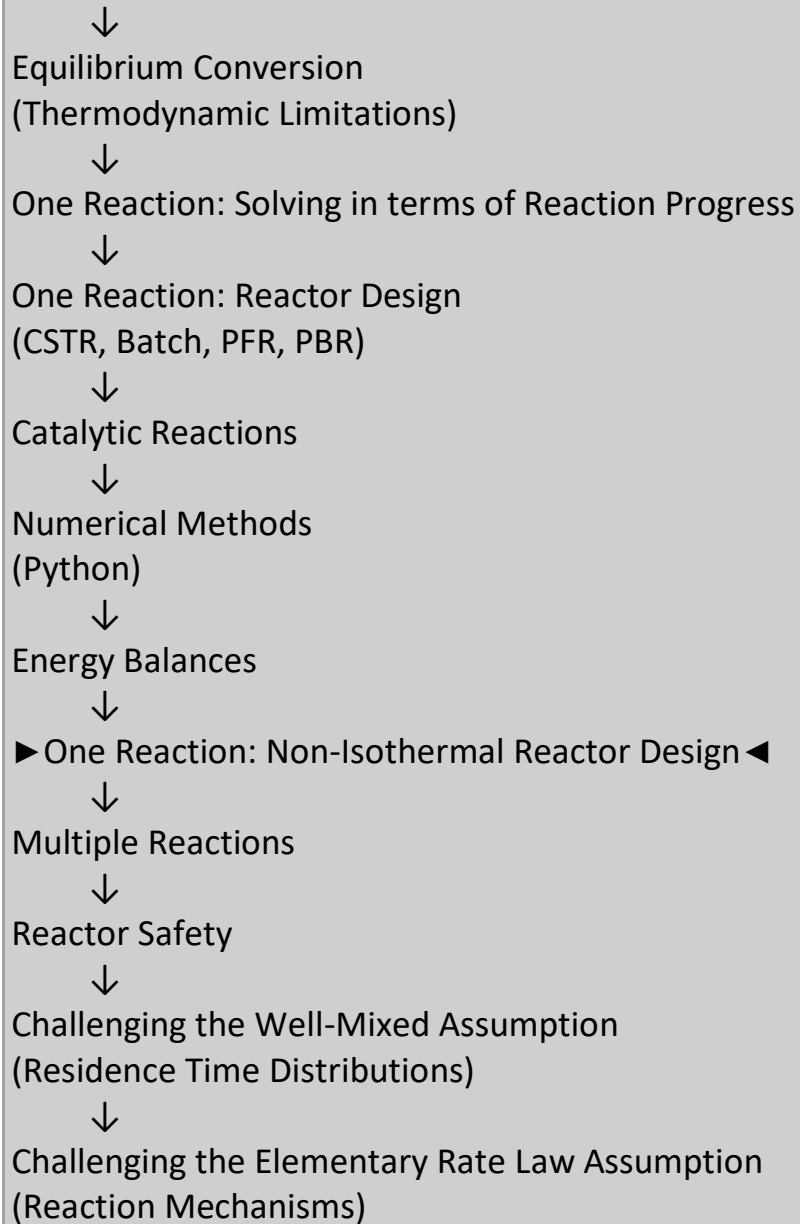
Graphical Reactor Design
(Inverse Rate/Levenspiel Plots)



Rate Laws
(Kinetics)



Equilibrium
(Reversible Reactions)



Learning Objectives

By the end of this lesson, you should be able to:

- Explain what an ignition-extinction curve represents.
- Predict how reactor temperature changes as feed temperature changes.
- Determine which steady state will be reached after a disturbance.
- Explain ignition and extinction temperatures.

- Describe how reactor history influences reactor operation.
- Explain why unstable steady states are rarely observed in practice.

Big Idea

Feed Temperature Changes



Steady States Shift



Reactor Moves to Nearest Stable Solution

What Does an Ignition-Extinction Curve Tell Us?

An ignition-extinction curve relates:

Feed Temperature



Reactor Temperature

for a non-isothermal CSTR. Rather than showing all possible steady states, the curve shows how the steady state the reactor is expected to reach as operating conditions are changed gradually.

Physical Interpretation

Specify Feed Temperature



Predict Reactor Temperature

Decision Guide: What Happens If I Increase Feed Temperature?

Suppose the reactor is operating on the lower-temperature branch.

Ask:

Increase Feed Temperature



Where Is the Next Available Steady State?

The reactor temperature will increase until it reaches the nearest stable operating point.

Typical Behavior

Increase Feed Temperature



Reactor Heats Slightly



Move to Nearby Stable Steady State until the ignition temperature is reached.

Decision Guide: What Happens If I Decrease Feed Temperature?

Suppose the reactor is operating on the upper-temperature branch.

Ask:

Decrease Feed Temperature



Where Is the Next Available Steady State?

The reactor cools until it reaches the nearest stable operating point.

Typical Behavior

Decrease Feed Temperature



Reactor Cools Slightly



Move to Nearby Stable Steady State until the extinction temperature is reached.

Multiple Steady States Do Not Mean Multiple Reactor Temperatures

When three intersections exist, students often conclude: The Reactor Operates At Three Temperatures.

This is incorrect.

Instead: Three Temperatures Are Possible but the reactor will operate at only one of them.

Key Question

Where Was the Reactor Operating Previously? The answer determines which steady state is reached after the operating conditions change.

Reactors Have Memory

Unlike many algebra problems, the operating state of a CSTR depends on its previous operating condition.

Example

Start on Lower Branch



Increase Feed Temperature



Remain on Lower Branch Until Ignition

Start on Upper Branch



Decrease Feed Temperature



Remain on Upper Branch Until Extinction

Key Insight

Current Conditions + Previous Operating State



Determine Reactor Response

The reactor does not simply choose among all possible steady states. Instead, it moves to the stable steady state that is closest to its current operating condition.

This behavior is known as **hysteresis**:

History of Reactor Operation



Influences Current Behavior

As a result, two reactors operating at the same feed temperature can sometimes be found at different reactor temperatures depending on how they reached that operating condition.

Why Is Ignition Important?

Near the ignition temperature, a very small change in feed temperature can cause a very large change in reactor temperature.

Physical Interpretation:

Small Increase in T_0



Loss of Lower Steady State



Jump to Upper Branch



Large Temperature Rise

The reactor temperature may increase dramatically even though the feed temperature changed only slightly.

Engineering Concern

Unexpected Temperature Rise



Possible Thermal Runaway

Decision Guide: Am I Near Ignition or Extinction?

If Operating Away From Ignition/Extinction

Small Change in Feed Temperature



Small Change in Reactor Temperature

If Operating Near Ignition/Extinction

Small Change in Feed Temperature



Large Change in Reactor Temperature

Quick Rule

Far From Ignition



Predictable Behavior

Near Ignition or Extinction



Potentially Dramatic Behavior

A reactor operating near the ignition or extinction temperature can exhibit very large temperature changes in response to very small changes in feed conditions.

Engineering Application: Reactor Startup

During startup:

Begin at Low Conversion



Gradually Heat Feed



Reach Ignition Temperature



Move to High-Conversion Branch

Knowing the ignition temperature allows engineers to determine the operating conditions required to reach the desired production rate.

Engineering Application: Reactor Safety

Ignition can cause a rapid increase in reactor temperature. Therefore:

Before Ignition Occurs



Cooling System Must Already Be Capable of Removing Additional Heat

Key Question: If Ignition Happens Today, Can My Cooling System Handle the Additional Heat Load?

Understanding ignition behavior is critical for preventing reactor runaway and maintaining safe operation.

Engineering Application: Reactor Shutdown

Extinction can be used as part of a reactor shutdown strategy.

Typical Procedure

Lower Feed Temperature



Reduce Reactor Temperature



Cross Extinction Point



Move to Low-Temperature Branch

Understanding the extinction temperature helps engineers predict how the reactor will respond during shutdown operations.

Summary and Key Takeaways

Key Ideas

- ✓ An ignition-extinction curve relates feed temperature to reactor temperature.
- ✓ Multiple steady states may exist for the same operating conditions.
- ✓ The reactor operates at the stable steady state closest to its previous operating condition.
- ✓ Small temperature changes can cause large reactor responses near ignition and extinction points.
- ✓ Ignition produces a jump to a higher-temperature operating state.
- ✓ Extinction produces a jump to a lower-temperature operating state.
- ✓ Knowledge of ignition and extinction temperatures is important for startup, shutdown, and reactor safety.

Most Important Idea

Multiple Steady States Exist



Reactor Does Not Choose Randomly



Moves to the Nearest Stable Steady State

When operating a non-isothermal CSTR, knowing where the reactor was previously operating is often just as important as knowing the current operating conditions.